

ECEN 428/722 - Lab Report

Lab Number: 2

Lab Title: Understanding the FPGA Design Cycle

Section Number: 602

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Date: 9/18/2025

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Objectives:

The aim of this lab is to use Vivado to:

1. Verify the functional correctness of RTL designs through behavioral simulation.
2. Explore synthesis and examine how pin, timing, and placement constraints influence device implementation on the FPGA.
3. Apply this process to understand the realization of FIR and IIR filters.

Design Process

The starting point was validating the provided FIR source files (`fir.v`, `multiply.v`) using the supplied testbench (`fir_tb.v`). Behavioral simulation confirmed that the filter produced the expected results.

After simulation, the Zybo constraint file was applied and we studied how different constraints—pin, timing, and placement—altered the placement of logic on the FPGA.

Instead of generating a free-running clock in the HDL, the design used a PLL generated via the Clocking Wizard IP. The 125 MHz on-board oscillator was converted to a 200 MHz clock output, which drove the filter logic.

Static timing analysis confirmed that the design mapped to the device correctly. In all cases the analysis reported **WNS > 0** and **TNS = 0**, meaning timing requirements were satisfied.

Results

- **Behavioral Simulation**

The simulation showed that the FIR filter output (`y`) responded correctly to the testbench input (`x`), producing results consistent with the expected delayed-input summation.

- **No Constraints**

Without constraints, placement was scattered and not optimized for timing. The design defaulted to the XOY0 region of the FPGA.

- **With Pin Constraints**

Once pin mappings were specified, the filter logic was placed closer to the I/O banks, reflecting the physical pin assignments.

- **With Pin and Timing Constraints**

Timing constraints caused Vivado to reorganize placement, shortening signal paths to reduce propagation delay.

- **With Pin, Timing, and Placement Constraints**

When placement constraints (pblock) were also added, the logic was confined to a defined region of the device.

- **With Only Placement Constraints**

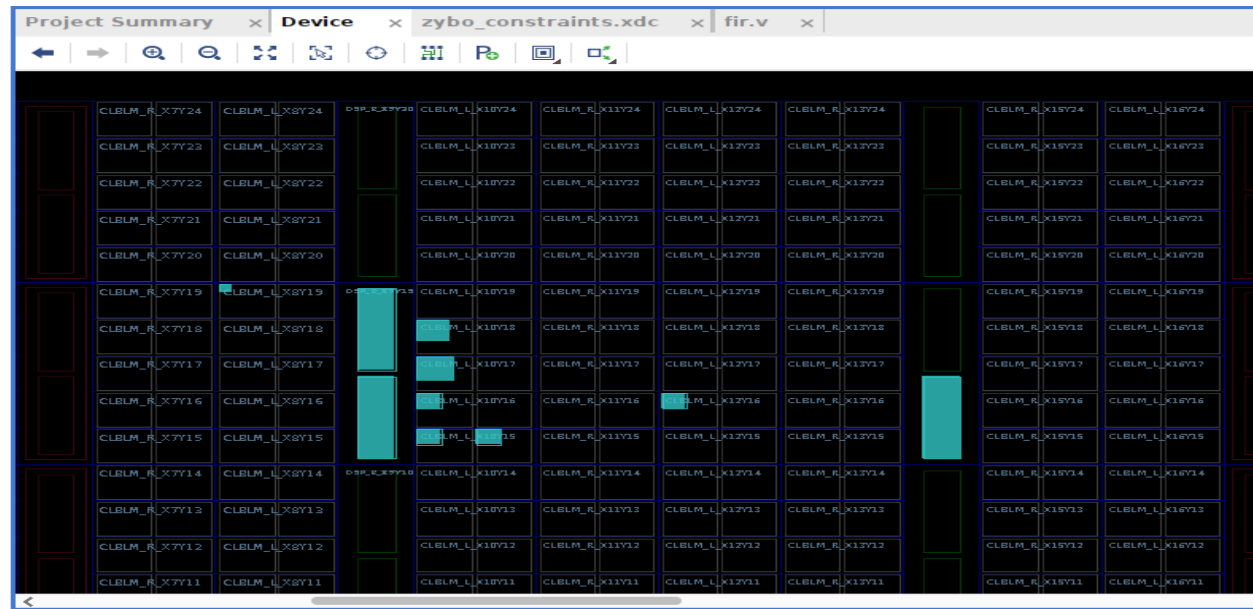
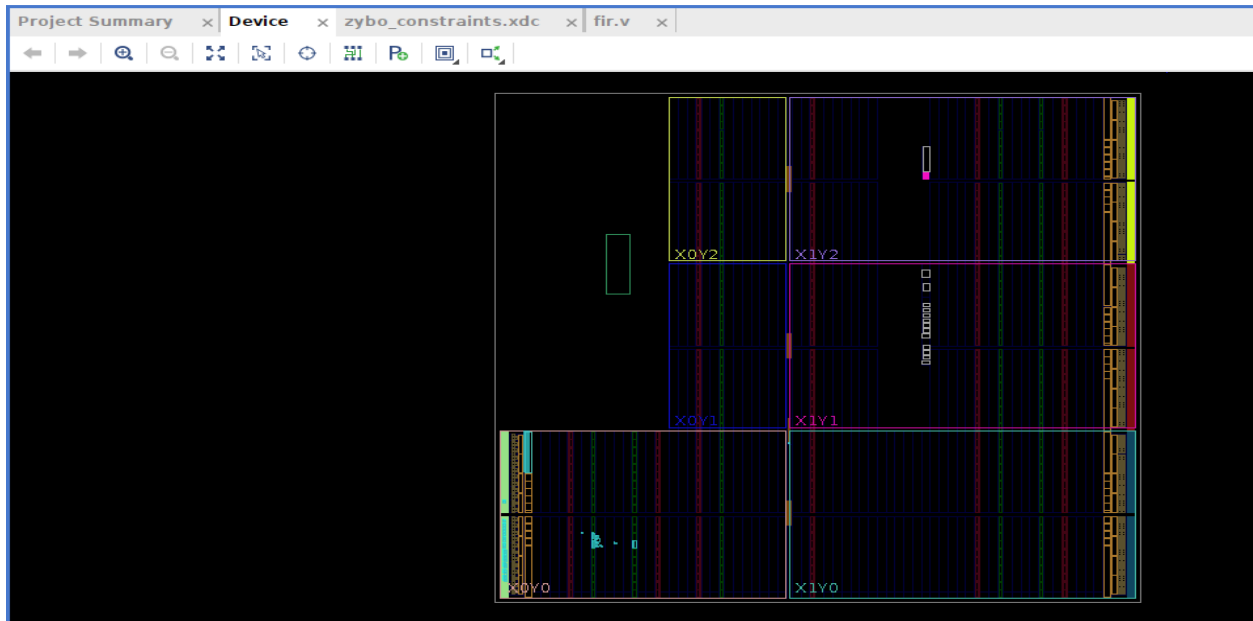
Placement alone did not optimize timing; the WNS degraded compared to runs with timing constraints applied.

Behavioral simulation:



The behavioral simulation for the FIR filter is simulated perfectly and the input and output signals are generated as expected.

A. The layout of the implemented diagram without any constraints



Design Timing Summary

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 3.753 ns	Worst Hold Slack (WHS): 0.221 ns	Worst Pulse Width Slack (WPWS): 2.000 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 8	Total Number of Endpoints: 8	Total Number of Endpoints: 22

All user specified timing constraints are met.

B. With pin-port constraints only



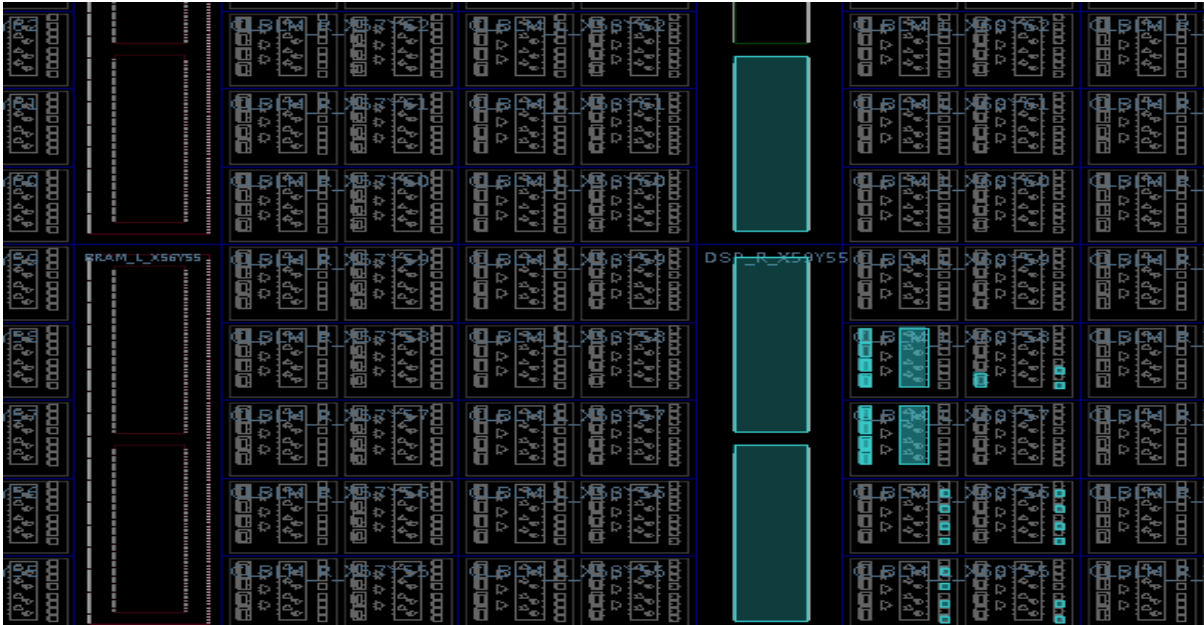
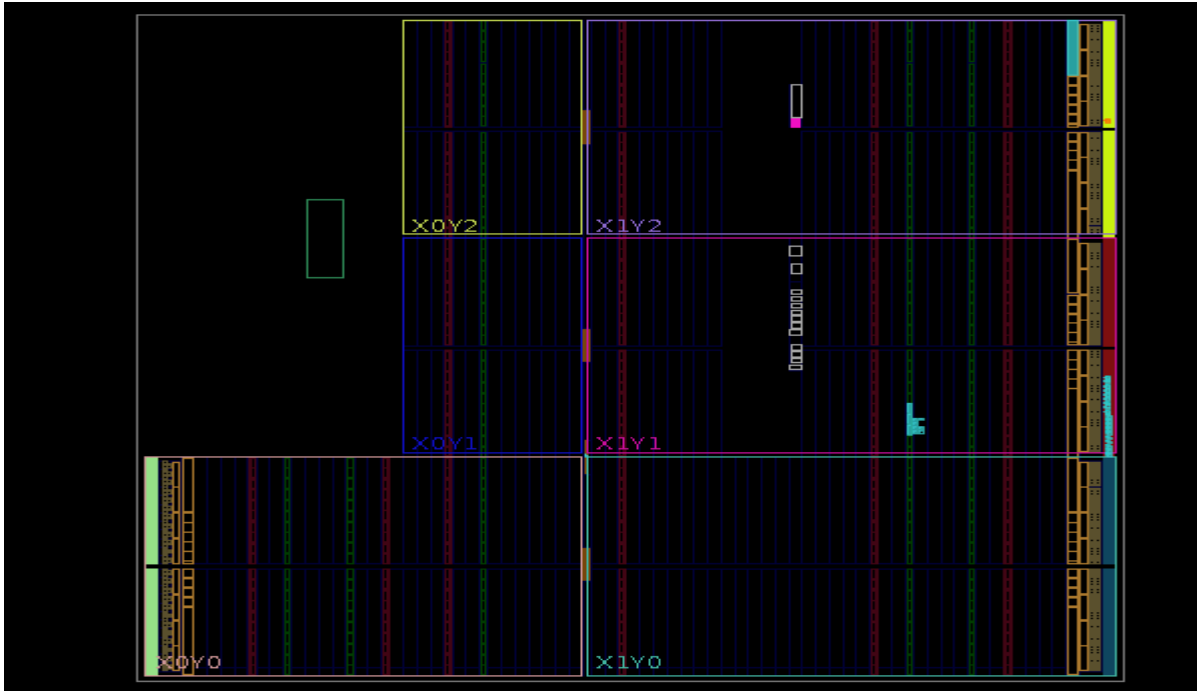
Project Summary x Device x zybo_constraints.xdc x fir.v x													
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Design Timing Summary

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 3.625 ns	Worst Hold Slack (WHS): 0.168 ns	Worst Pulse Width Slack (WPWS): 2.000 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 8	Total Number of Endpoints: 8	Total Number of Endpoints: 22

All user specified timing constraints are met.

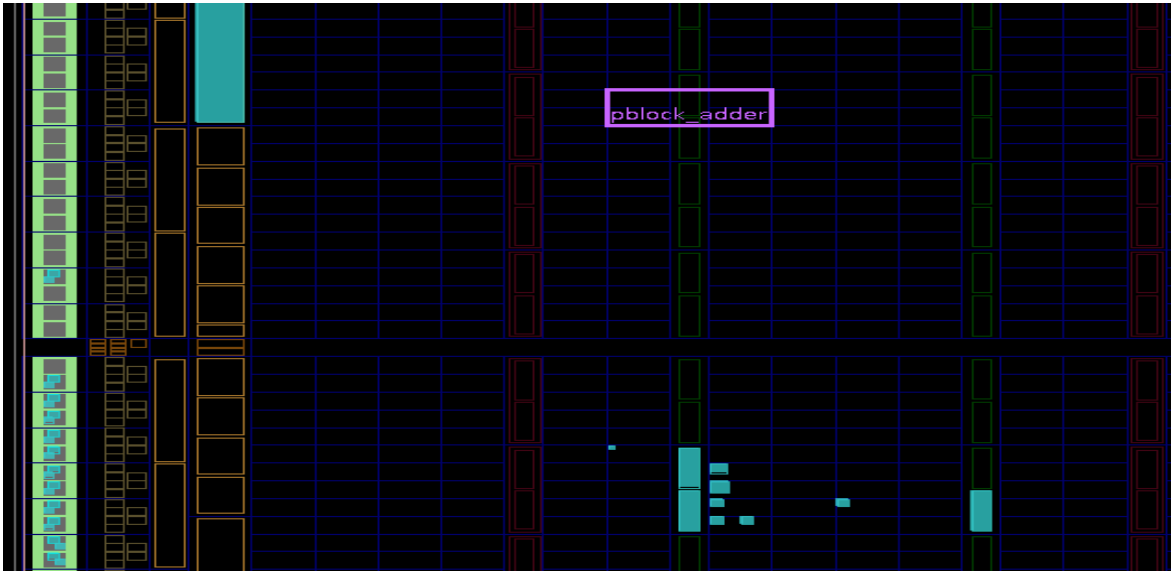
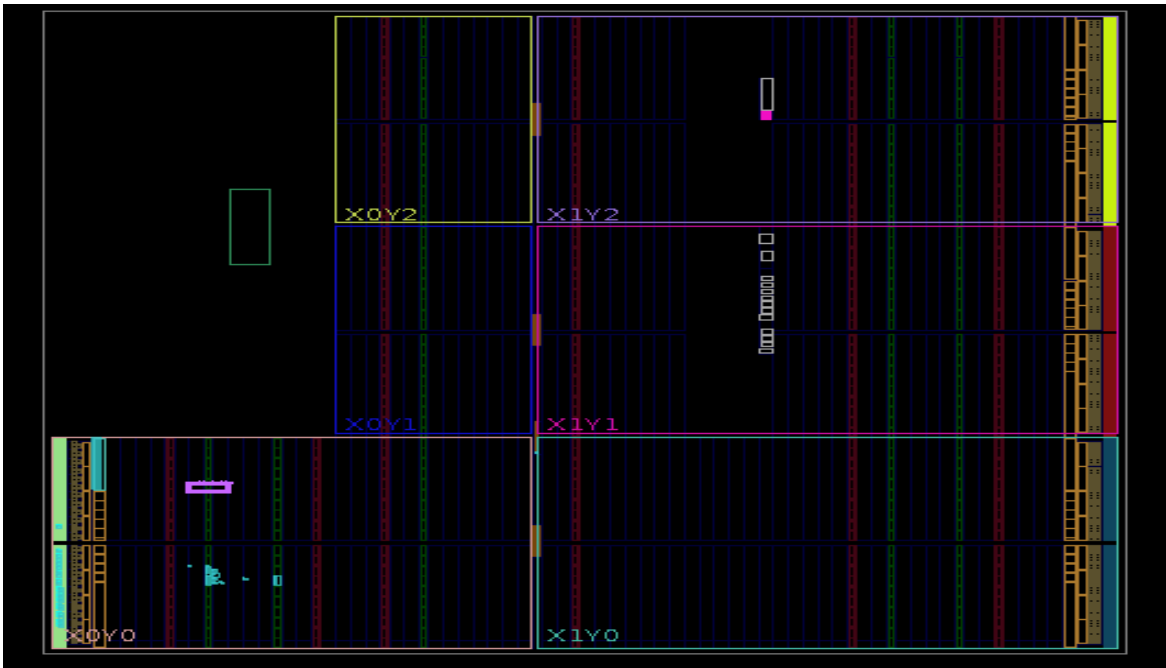
C. With Timing constraints only



Design Timing Summary

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 3.760 ns	Worst Hold Slack (WHS): 0.185 ns	Worst Pulse Width Slack (WPWS): 2.000 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 8	Total Number of Endpoints: 8	Total Number of Endpoints: 22
All user specified timing constraints are met.		

D. With placement constraints only

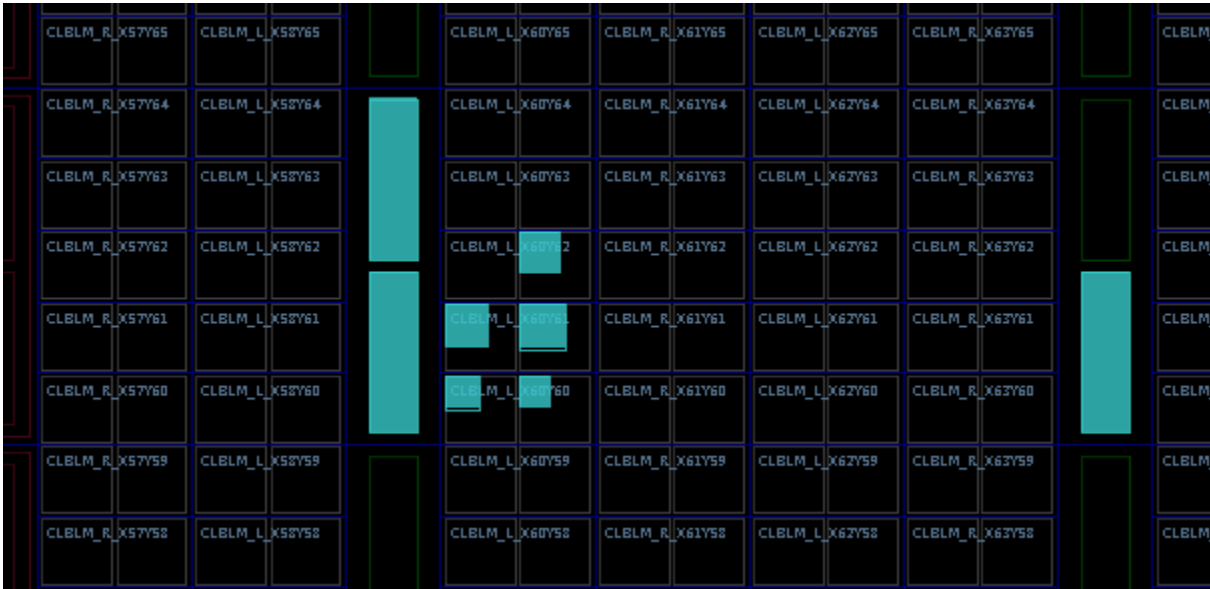
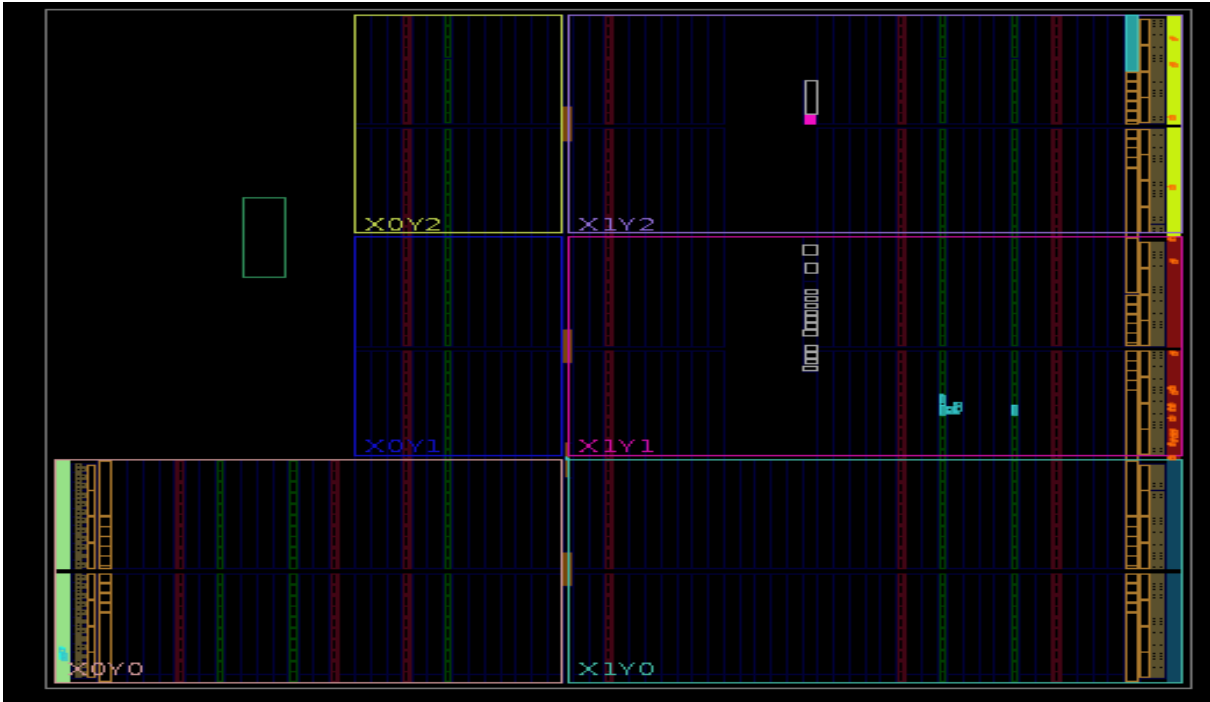


Design Timing Summary

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 3.753 ns	Worst Hold Slack (WHS): 0.221 ns	Worst Pulse Width Slack (WPWS): 2.000 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 8	Total Number of Endpoints: 8	Total Number of Endpoints: 22

All user specified timing constraints are met.

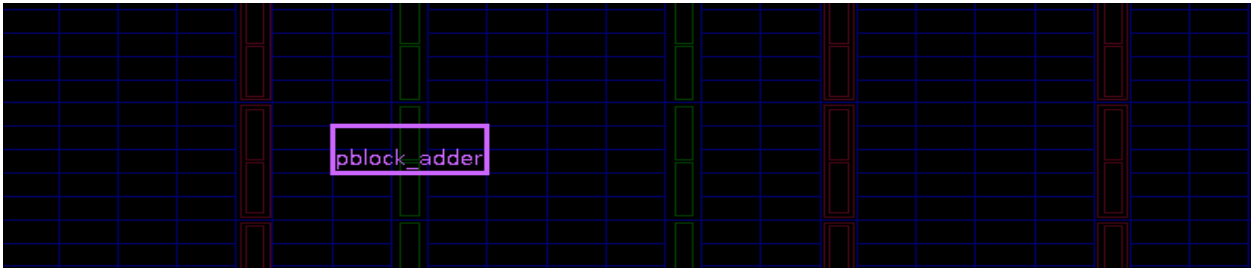
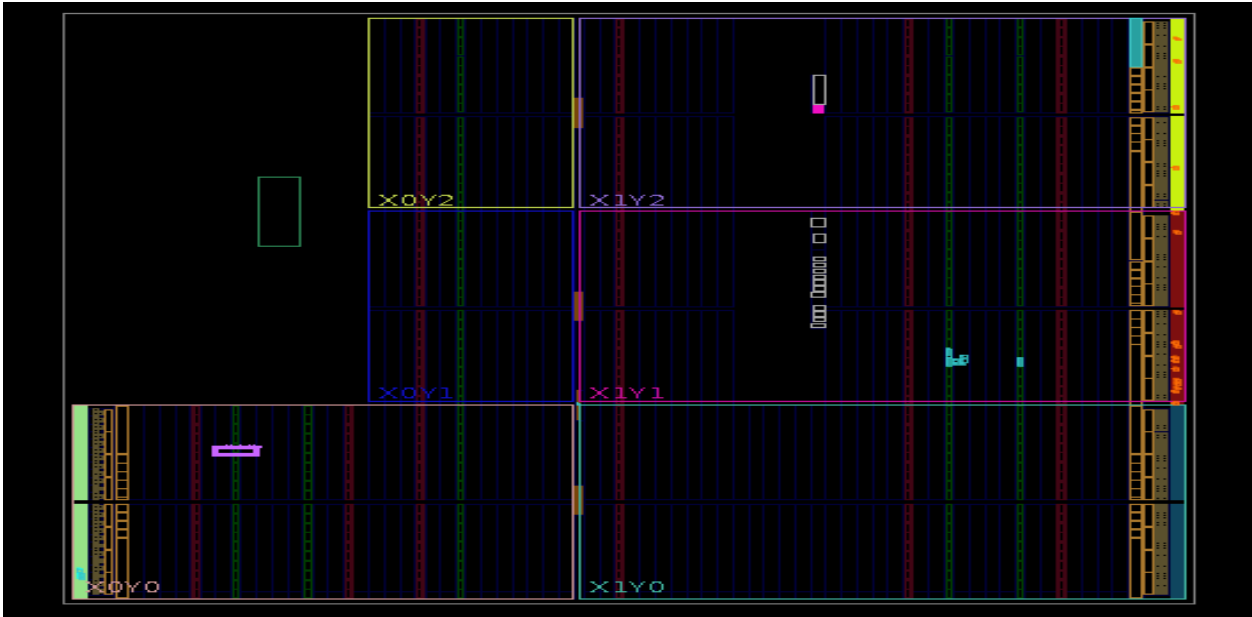
E. With pin and timing constraints



Design Timing Summary

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 3.582 ns	Worst Hold Slack (WHS): 0.168 ns	Worst Pulse Width Slack (WPWS): 2.000 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 8	Total Number of Endpoints: 8	Total Number of Endpoints: 22
All user specified timing constraints are met.		

F. With all constraints



CLBLM_R_X57Y65	CLBLM_L_X54Y65	CLBLM_R_X50Y65	CLBLM_R_X57Y65	CLBLM_L_X50Y65	CLBLM_L_X60Y65	CLBLM_R_X61Y65	CLBLM_L_X62Y65	CLBLM_R_X67Y65	CLBLM_L_X60Y65	CLBLM_R_X65Y65	CLBLM_L_X66Y65
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Design Timing Summary

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 3.582 ns	Worst Hold Slack (WHS): 0.168 ns	Worst Pulse Width Slack (WPWS): 2.000 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 8	Total Number of Endpoints: 8	Total Number of Endpoints: 22
All user specified timing constraints are met.		

A. No Constraints

The design elements are scattered without optimization (defaulted to XOY0 region). Since no timing information is given, slack values are not defined, and the tool does not optimize for path delays.

B. With Pin Constraints

Placement shifts closer to the I/O banks as per pin mappings. Slack is still not optimized because timing constraints are missing, though functionality is preserved.

C. With Timing Constraints Only

Vivado uses the defined clock to reorganize placement and shorten critical paths. Timing analysis shows $WNS > 0$ and $TNS = 0$, confirming successful timing closure.

D. With Placement Constraints Only

Logic is forced into a fixed region (pblock), but without timing guidance slack degrades. WNS is smaller, and timing is less favorable compared to runs with proper timing constraints.

E. With Pin and Timing Constraints

Both physical placement (pins) and timing information are considered. The design shows improved path optimization, with positive slack ($WNS > 0$, $TNS = 0$) achieved.

F. With All Constraints

Combining pin, timing, and placement constraints produces a compact, well-structured layout. Timing analysis confirms the best results, with positive WNS and zero TNS, ensuring robust closure.

Conclusion:

This exercise demonstrated the impact of different constraint types on FPGA design implementation. Pin, timing, and placement constraints collectively guide Vivado toward optimized results. Comparing across scenarios made clear the contribution of each constraint toward achieving both correct functionality and efficient timing closure.